



**SEMESTER 1
2021-2022**

**CS264FZ
Software Design**

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Time allowed: 2 hours

Answer all 4 questions

Instructions

	Yes	No
Log Books Allowed	☐	X
Formula Tables Allowed	☐	X
Other Allowed (<i>enter details</i>)	☐	X

General (*enter details*)

- [25 marks]**
- 1** (a) What is the relationship and difference between the Singleton and Flyweight patterns? [5 marks]
- (b) How many types of adapters does the Adapter pattern support? Discuss the differences between them briefly. [5 marks]
- (c) Explain the idea behind the Abstract Factory pattern. [5 marks]
- (d) In the Facade pattern, is it possible to create more than one facade for a complex subsystem? Provide a justification for your answer. [5 marks]
- (e) The Bridge pattern is similar to the State pattern. Is the statement correct? Why? [5 marks]
- [25 marks]**
- 2** (a) What are the key advantages associated with the Command pattern? [4 marks]
- (b) In the Chain-of-Responsibility pattern, how can you handle the scenario where the request is not handled when you have reached the end of a chain? [6 marks]
- (c) In the Mediator pattern, suppose that you are reducing the number of interconnections among various objects. What key benefits will you achieve from the reduction? [5 marks]
- (d) In most implementations of the Memento pattern, the Memento class does not have a public setter method. What is the reason behind this? [6 marks]
- (e) How can we use the Iterator pattern? Please give an example. [4 marks]

[25 marks]

- 3** (a) Suppose that a person is ready to go abroad. His friend asks him to buy a product. Please use the Proxy pattern to describe the scene.

(1) Write a Person interface with a buyProduct method.

[5 marks]

(2) Write two classes of ProxyBuyPerson and RealBuyPerson inheriting from the Person interface. Class ProxyBuyPerson should have a constructor and an object of RealBuyPerson as its attribute. It should also overload the buyProduct method to call the buyProduct method on the RealBuyPerson attribute.

[10 marks]

(3) Class RealBuyPerson should have a constructor to input a product's name as its attribute (productName). It should also overload the buyProduct method to output a message "I need *(product name)".

[10 marks]

[25 marks]

- 4 (a) There is a weather website that publishes some real-time weather statistics (temperature and humidity). There are two subscribers to this website. Whenever the weather information on the website changes, it will be automatically pushed to the subscribers. Please use the Observer pattern to describe the scene.
- (1) Write a Subject interface with three methods: registerObserver(Observer o), removeObserver(Observer o) and notifyObservers(). [5 marks]
 - (2) Write a WeatherData class that implements the Subject interface. It has two float-typed attributes to describe the temperature and humidity information. It has an attribute of ArrayList<Observer> (observers) and three methods: getTemperature(), getHumidity() and setData(float temperature, float humidity). [5 marks]
 - (3) Write an Observer interface with method update(float temperature, float humidity). [5 marks]
 - (4) Write a UserA class and a UserB class. These two classes implement the Observer interface. They have two float-typed attributes of temperature and humidity. They have a display() method to print weather information "temperature=xxx, humidity=xxx". [10 marks]